Midterm Sample: Solutions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

1. (5 pts)

   (a) (1 pt) *Is f a valid function?* **Yes**: a function may map two values of x to the same value of y; what it cannot do is map one x to two values of y.

   (b) (1 pt) $A \cup B =$ Set of all integers.

   (c) (1 pt) $A \cup B = A \qquad A \cap B = B$

   (d) (1 pt) No: since the function is not one-to-one (it is not monotonic), the inverse does not exist.

   (e) (1 pt) $(0, 2)$

2. (5 pts) $A = I - X(X^T X)^{-1} X^T$

   (a) (3 pts) Say the dimension of X is $m \times n$. Then the dimension of $X^T_{n\times m} X_{m\times n}$ is $n \times n$, so the dimension of $(X^T X)^{-1}$ is also $n \times n$. This implies that the dimension of $X_{m\times n}(X^T X)^{-1}_{n\times n} X^T_{n\times m}$ is $m \times m$. Hence A is square (as is $X^T X$), but X need not be square.

   (b) (2 pts)

$$
\begin{aligned}
AA &= (I - X(X^T X)^{-1} X^T)(I - X(X^T X)^{-1} X^T) \\
&= I - X(X^T X)^{-1} X^T - X(X^T X)^{-1} X^T + X \underbrace{(X^T X)^{-1} X^T X}_{I} (X^T X)^{-1} X^T \\
&= I - X(X^T X)^{-1} X^T = A
\end{aligned}
$$

3. (10 pts)

   (a) (2 pts) Setting supply equal to demand in each market:

$$30 + p_e = 100 - 2p_e + p_g \implies 3p_e - p_g = 70$$

$$20 + p_g = 50 + p_e - p_g \implies -p_e + 2p_g = 30$$

(b) (2 pts)

$$A = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}_{2\times 2} \qquad x = \begin{bmatrix} p_e \\ p_g \end{bmatrix}_{2\times 1} \qquad b = \begin{bmatrix} 70 \\ 30 \end{bmatrix}_{2\times 1}$$

(c) (2 pts) The cofactors are $|C_{11}| = 2$, $|C_{12}| = 1$, $|C_{21}| = 1$, $|C_{22}| = 3$, so

$$adjA = C^T = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

(d) (1 pt) $|A| = 3 \times 2 - (-1) \times (-1) = 5 \neq 0$, so A is nonsingular.

(e) (1 pt) $A^{-1}Ax = A^{-1}b \implies x^* = A^{-1}b$

(f) (2 pts)

$$x^* = A^{-1}b = \frac{1}{|A|} adjA\, b = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 70 \\ 30 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 170 \\ 160 \end{bmatrix} = \begin{bmatrix} 34 \\ 32 \end{bmatrix}$$

So $p_e^* = 34$ and $p_g^* = 32$ (and, substituting back, $q_e^* = 64$, $q_g^* = 52$).